

Layered Kagome Niobium Halides and Chalcogenide-Halides: A Platform for Optical, Thermodynamic, Piezoelectric, and Flexoelectric Responses

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Abstract

Layered kagome niobium halides and chalcogenide-halides offer a chemically tunable platform in which breathing-kagome geometry, metal-cluster orbitals, structural polarity, and lattice flexibility can be integrated to realize anisotropic optical, thermodynamic and electromechanical functionalities. Here, we systematically investigate non-Janus Nb_3X_8 ($\text{X} = \text{Cl}, \text{Br}, \text{I}$) and Janus M_3YZ_7 ($\text{M} = \text{Nb}, \text{Ta}$; $\text{Y} = \text{S}, \text{Se}, \text{Te}$; $\text{Z} = \text{Cl}, \text{Br}, \text{I}$) monolayers using first-principles density functional theory calculations and molecular dynamics simulations based on trained moment tensor potentials combined with a charge–dipole model. The calculated optical spectra reveal pronounced anisotropy between the in-plane ($E\parallel b$) and out-of-plane ($E\parallel c$) polarization directions, confirming that light–matter interaction is dominated by the in-plane Nb-based breathing-kagome network. For Nb_3X_8 , the optical and thermodynamic response follows the order $\text{Nb}_3\text{I}_8 > \text{Nb}_3\text{Br}_8 > \text{Nb}_3\text{Cl}_8$, which is attributed to increasing halogen polarizability, stronger spin–orbit coupling, and enhanced Nb- d /halogen- p hybridization from Cl to I. The absorption coefficient reaches the order of 10^6 cm^{-1} , indicating strong visible and near-ultraviolet absorption. In Janus Nb_3YZ_7 compounds, Nb_3SeI_7 exhibits the strongest peak optical response, whereas Nb_3TeI_7 provides a broader absorption and dielectric window because of diffuse Te- p states and spin–orbit-induced band mixing. The electromechanical response shows a clear composition-dependent stiffness and polarization trend. Among the non-Janus halides, Nb_3Cl_8 has the largest Young’s modulus and is approximately 1.10 and 1.27 times stiffer than Nb_3Br_8 and Nb_3I_8 , respectively, demonstrating progressive lattice softening from Cl to Br and I. The trained MTPs reproduce the DFT mechanical trends with good accuracy, confirming their reliability for large-scale elastic deformation simulations. The polarization–strain response shows that Janus M_3YZ_7 monolayers possess stronger piezoelectricity than non-Janus Nb_3X_8 , with the largest response approaching $\sim 0.10 \text{ nC/m}$ for highly polar Janus systems. This enhancement originates from broken out-of-plane mirror symmetry, asymmetric charge redistribution, local electric-field variation, and mechanically soft bonding networks. Under bending deformation, the calculated flexoelectric coefficients reach the order of $\sim 0.3\text{--}0.8 \text{ nC/m}$, with Nb-based Janus I-containing compounds showing the strongest strain-gradient-induced polarization.

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1. Introduction

The search for multifunctional two-dimensional (2D) materials has moved beyond graphene and conventional transition-metal dichalcogenides (TMDs) toward crystal architectures in which lattice geometry, chemical composition, and reduced dimensionality act together to generate emergent properties [1, 2, 3]. Among these systems, kagome and breathing-kagome lattices have attracted particular attention because their networks of corner-sharing triangles can host flat electronic bands, Dirac-like dispersions, van Hove singularities, localized molecular-orbital states, and strong correlation effects [4, 5]. In a breathing-kagome lattice, the triangular units are inequivalent, producing alternating small and large triangles. This structural distortion modifies the electronic hopping pathways and can generate narrow bands associated with metal-cluster molecular orbitals. Therefore, breathing-kagome materials provide a fertile platform for exploring flat-band physics, Mott insulating behaviour, magnetism, topology, optical anisotropy, and electromechanical coupling [6, 7]. Layered niobium-based kagome halides and chalcogenide-halides are especially important within this class of materials. The non-Janus binary halides Nb_3X_8 ($\text{X} = \text{Cl}, \text{Br}, \text{I}$) and Janus chalcogenide-halides M_3YZ_7 ($\text{M} = \text{Nb}, \text{Ta}$; $\text{Y} = \text{S}, \text{Se}, \text{Te}$; $\text{Z} = \text{Cl}, \text{Br}, \text{I}$) contain triangular metal clusters arranged in kagome or breathing-kagome frameworks. In these compounds, the electronic states near the Fermi level are strongly influenced by Nb d -orbitals and cluster molecular orbitals formed by metal trimers. This cluster-based electronic structure distinguishes them from ordinary TMDs, where the band-edge physics is mainly governed by transition-metal d -orbitals at the K and K' valleys. As a result, layered niobium kagome compounds provide a chemically tunable platform where electronic correlation, spin-orbit coupling, optical response, mechanical flexibility, and polarization-related effects can be investigated in a unified materials family [8, 9, 10].

The binary niobium halides Nb_3X_8 have been widely studied because of their layered van der Waals (vdWs) structure and breathing-kagome arrangement of Nb trimers. The successful exfoliation and observation of topological flat bands in Nb_3Cl_8 stimulated broad interest in this family [11]. Theoretical studies further suggested that monolayer Nb_3X_8 systems are close to the strong-correlation limit and can be described using effective Hubbard-type models [12]. Among them, Nb_3Cl_8 is a prototypical layered cluster Mott insulator, where charge localization, Nb-trimer molecular orbitals, and magnetic-nonmagnetic transitions are closely connected [6, 13]. Pressure-dependent studies have also shown that the insulating state can be suppressed and that metallic behaviour may emerge under compression [14]. In Nb_3Br_8 , angle-resolved photoemission spectroscopy has revealed flat and weakly dispersive bands associated with Nb d -orbital states in the breathing-kagome plane [15]. Nb_3I_8 is also noteworthy because iodine introduces stronger spin-orbit coupling, and its electronic and magnetic properties are sensitive to layer thickness, strain, stacking order, and interlayer coupling [16]. These results demonstrate that chemical substitution from Cl to Br and I provides an effective route for tuning flat-band electronic structure, lattice softness, and spin-orbit-driven phenomena. In addition to their electronic properties, the optical response of Nb_3X_8 compounds has attracted increasing attention. First-principles studies predicted that monolayer Nb_3X_8 materials are semiconducting and exhibit strong visible and infrared light absorption [8]. More recently, Nb_3Cl_8 was reported to show strong optical anisotropy, with a much

larger in-plane refractive index, dielectric response, and absorption coefficient than the out-of-plane components [17]. Such anisotropy is a natural consequence of the layered structure and the dominant in-plane Nb-based kagome electronic network. However, a systematic comparison of the full optical response, including refractive index, extinction coefficient, absorption coefficient, reflectivity, and dielectric components, across Nb_3Cl_8 , Nb_3Br_8 , and Nb_3I_8 remains important for understanding how halogen substitution controls light–matter interaction. The mechanical properties of these materials are equally important because mechanical flexibility strongly influences piezoelectric and flexoelectric responses in 2D systems. Mortazavi *et al.* [9] investigated the mechanical behaviour of Nb_3X_8 monolayers and showed that the Young’s modulus and tensile strength decrease with increasing halogen atomic weight. This trend originates from longer Nb–X bonds and weaker effective force constants when Cl is replaced by Br and I. Such composition-dependent lattice softening is highly relevant for electromechanical applications because a softer lattice can produce larger internal atomic displacement under strain. Nevertheless, the relationship between mechanical softness, strain-induced polarization, and curvature-induced polarization in Nb_3X_8 and related Janus kagome systems has not been fully clarified.

Ternary chalcogenide-halides, including M_3YZ_7 , have recently emerged as an important extension of the niobium kagome family. Partial replacement of halogen atoms by chalcogen atoms creates noncentrosymmetric or Janus-like structures, leading to intrinsic polarity and broken out-of-plane mirror symmetry [10, 18]. This structural asymmetry is important because it enables nonlinear optical effects, vertical charge separation, piezoelectricity, and polar electronic behaviour. For example, Nb_3SeI_7 has been experimentally shown to exhibit polarization-independent second-harmonic generation due to its noncentrosymmetric breathing-kagome structure [19]. Recent studies also demonstrated out-of-plane piezoelectricity in Janus Nb_3SeI_7 , highlighting its potential for piezoelectric and piezophototronic applications [20]. Similarly, Nb_3TeI_7 has been reported to show flat-band electronic bipolarity, built-in polarity, and surface-dependent electronic behaviour, making it attractive for polar electronics and asymmetric device concepts [21]. More broadly, large second-harmonic generation and linear electro-optic responses have been predicted in the Nb_3MX_7 family, demonstrating that the combination of kagome geometry, cluster orbitals, and broken inversion symmetry can produce strong nonlinear optical coefficients [22]. Importantly, the M_3QX_7 family is not only of theoretical interest but also has strong experimental relevance. Several bulk members of this family, where $M = \text{Nb}, \text{Ta}$, $Q = \text{S}, \text{Se}, \text{Te}$, and $X = \text{Cl}, \text{Br}, \text{I}$, have already been synthesized and structurally characterized. Experimental reports show that these compounds can crystallize in different symmetry classes, including the trigonal $P3m1$ and $P\bar{3}m1$ space groups and the hexagonal $P6_3mc$ space group. For example, Nb_3SBr_7 and Nb_3TeCl_7 have been reported in trigonal structures, whereas Ta_3SeI_7 and Ta_3TeI_7 adopt hexagonal structures [23, 24, 25]. In addition, Nb_3TeI_7 is particularly notable because it exhibits polymorphism, with both trigonal and hexagonal structural variants reported experimentally [18]. These compounds are commonly prepared using solid-state reaction routes from stoichiometric mixtures of transition-metal, chalcogen, and halogen precursors sealed in evacuated quartz ampoules. Recent synthesis of high-quality Nb_3TeCl_7 single crystals further demonstrates the experimental accessibility of this family [26]. These experimental realizations strongly support the feasibility of synthesizing additional M_3QX_7 compounds and motivate systematic investigations of their optical, mechanical, piezoelectric, and flexoelectric properties.

Piezoelectricity and flexoelectricity provide additional routes for utilizing these layered kagome monolayers in electromechanical devices. Conventional 2D piezoelectrics, such as monolayer h-

BN and 2H-TMDs including MoS₂, WS₂, MoSe₂, and WSe₂, exhibit in-plane piezoelectricity because inversion symmetry is broken in the monolayer limit [27, 28]. However, their out-of-plane piezoelectric response is usually absent or weak because of mirror symmetry with respect to the basal plane. In contrast, Janus monolayers naturally break out-of-plane mirror symmetry, allowing vertical dipoles and enhanced electromechanical coupling. This makes M₃YZ₇ kagome monolayers particularly attractive because they combine Janus polarity with breathing-kagome electronic structure. In parallel, flexoelectricity, which couples polarization to strain gradients, can occur even in centrosymmetric systems and becomes especially important in 2D materials under bending deformation [29, 30]. Previous studies have shown that bending-induced flexoelectricity can be significant in graphene, h-BN, and TMDs [31]. However, the flexoelectric response of layered kagome niobium halides and chalcogenide-halides remains largely unexplored. Despite these advances, a unified understanding of the optical, thermodynamic, mechanical, piezoelectric, and flexoelectric properties of Nb₃X₈ and M₃YZ₇ layered kagome compounds is still lacking. Previous studies have mainly focused on selected electronic, magnetic, nonlinear optical, or mechanical properties of individual members of these families. A systematic comparison across non-Janus halides and Janus chalcogenide-halides is essential for revealing how halogen/chalcogen substitution, atomic polarizability, spin-orbit coupling, bonding strength, lattice softness, and structural polarity govern their multifunctional behaviour. Furthermore, large-scale molecular dynamics simulations require accurate interatomic potentials capable of reproducing first-principles mechanical and polarization responses.

In the present work, we systematically investigate the phonon dispersion, optical properties, pressure-dependent thermodynamic behaviour, mechanical response, piezoelectricity, and flexoelectricity of Nb₃X₈ ($X = \text{Cl, Br, I}$) and M₃YZ₇ ($M = \text{Nb, Ta}; Y = \text{S, Se, Te}; Z = \text{Cl, Br, I}$) layered kagome monolayers. The phonon, optical, and thermodynamic properties are evaluated using first-principles density functional theory calculations, whereas the mechanical, piezoelectric, and flexoelectric responses are studied using classical molecular dynamics simulations based on trained moment tensor potentials combined with a charge-dipole (QP) model. This combined approach provides a unified description of dynamical stability, anisotropic light-matter interaction, pressure-tunable thermal behaviour, strain-induced polarization, and bending-induced flexoelectricity. The results establish layered kagome niobium halides and chalcogenide-halides as promising multifunctional 2D materials for anisotropic optoelectronics, photodetectors, nonlinear optics, piezotronic devices, flexible electromechanical systems, and vdWs heterostructure engineering.

2. Computational methodology

First-principles calculations for different niobium-based kagome materials, including Nb₃X₈ ($X = \text{Cl, Br, I}$) and M₃YZ₇ ($M = \text{Nb, Ta}; Y = \text{S, Se, Te}; Z = \text{Cl, Br, I}$), were carried out within the framework of DFT using the Vienna ab initio Simulation Package (VASP) [32, 33]. The electron-ion interactions were described by the projector augmented-wave (PAW) method [34, 35], while the exchange-correlation energy was treated using the generalized gradient approximation (GGA) in the Perdew-Burke-Ernzerhof (PBE) [36]. The plane-wave kinetic energy cutoff was set to 520 eV, and the Brillouin zone was sampled using a Γ -centered $20 \times 20 \times 1$ k -point mesh. The self-consistent electronic convergence criterion was fixed at 10^{-8} eV. During structural optimization, the atomic positions and lattice parameters were relaxed until the Hellmann-Feynman forces were smaller than 10^{-2} eV $\text{\AA}^{-1}$ for volume relaxation and 10^{-3} eV $\text{\AA}^{-1}$ for ionic relaxation. To avoid

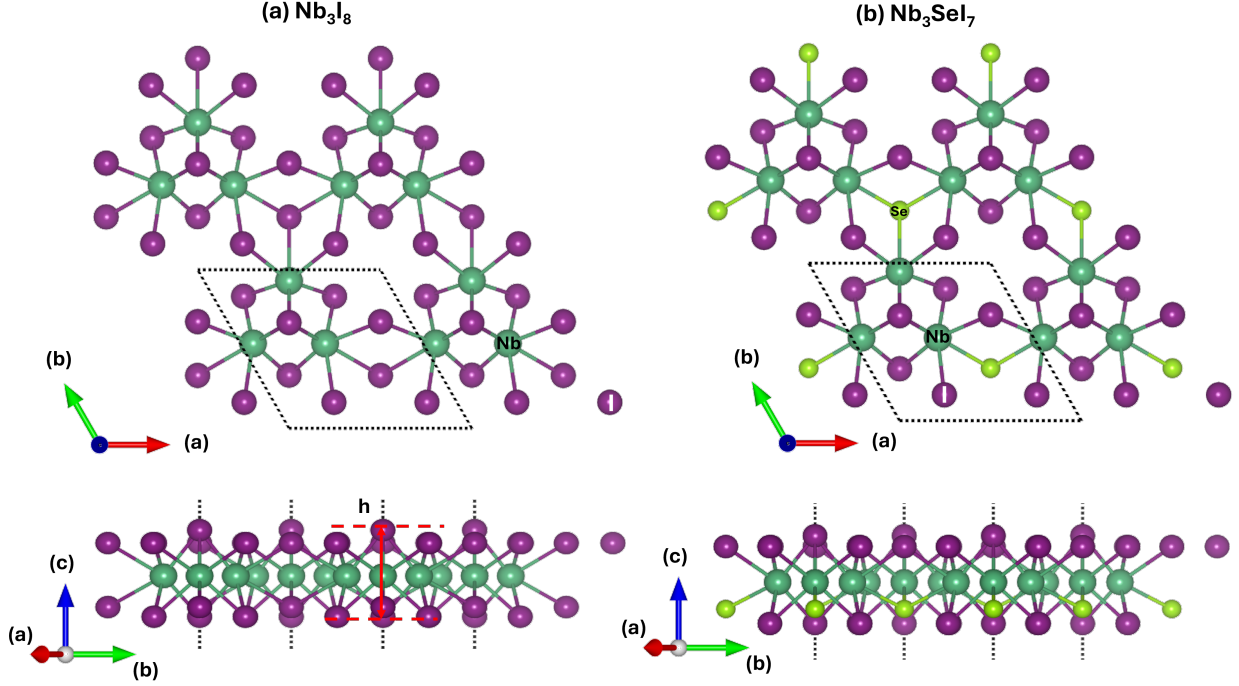


Fig. 1. Top and side views of representative layered kagome monolayers: (a) non-Janus Nb_3I_8 and (b) Janus Nb_3SeI_7 . The dashed rhombus denotes the primitive unit cell, and h represents the vertical distance between the outermost atomic planes used for estimating the vdW-corrected monolayer thickness.

artificial interactions between periodic images along the out-of-plane direction, a vacuum spacing of 25 Å was introduced. The dynamical stability of the optimized monolayers was examined by calculating the phonon dispersion spectra using the finite-displacement method [37, 38].

The distinct optical properties are evaluated such as refractive index $n(\omega)$, extinction coefficient $k(\omega)$, absorption coefficient $\alpha(\omega)$, reflectivity $R(\omega)$, and the real and imaginary parts of the dielectric function, $\varepsilon_1(\omega)$ and $\varepsilon_2(\omega)$, respectively. The optical response of a niobium-based kagome compounds can be described using the complex dielectric function, which determines how the material interacts with an incident electromagnetic wave. The complex dielectric function is expressed as

$$\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega), \quad (1)$$

where $\varepsilon_1(\omega)$ and $\varepsilon_2(\omega)$ are the real and imaginary parts of the dielectric function, respectively. The real part, $\varepsilon_1(\omega)$, represents the ability of the material to become polarized under the influence of an external electromagnetic field. It is therefore associated with dispersion and the storage of electromagnetic energy inside the material. In contrast, the imaginary part, $\varepsilon_2(\omega)$, describes the energy loss associated with electronic transitions and absorption processes. Thus, $\varepsilon_2(\omega)$ provides information about the ability of the material to absorb electromagnetic radiation and dissipate energy in the form of heat [39].

The complex refractive index is also related to the dielectric function and can be written as

$$\tilde{n}(\omega) = n(\omega) + ik(\omega), \quad (2)$$

where $n(\omega)$ is the refractive index and $k(\omega)$ is the extinction coefficient. According to Maxwell's equations, the complex dielectric function is related to the complex refractive index by

$$\varepsilon(\omega) = [n(\omega) + ik(\omega)]^2. \quad (3)$$

Expanding Eq. (3), the real and imaginary parts of the dielectric function can be expressed as

$$\varepsilon_1(\omega) = n^2(\omega) - k^2(\omega), \quad (4)$$

$$\varepsilon_2(\omega) = 2n(\omega)k(\omega) \quad (5)$$

From these relations, $n(\omega)$ and $k(\omega)$ can be obtained as

$$n(\omega) = \left[\frac{\sqrt{\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega)} + \varepsilon_1(\omega)}{2} \right]^{1/2}, \quad (6)$$

$$k(\omega) = \left[\frac{\sqrt{\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega)} - \varepsilon_1(\omega)}{2} \right]^{1/2}. \quad (7)$$

The refractive index $n(\omega)$ describes the phase velocity of light inside the material. A larger value of $n(\omega)$ indicates that light propagates more slowly through the material compared with its propagation in vacuum. The extinction coefficient $k(\omega)$ represents the attenuation of light inside the material due to absorption and scattering processes. Therefore, $k(\omega)$ is closely related to the energy loss of the electromagnetic wave during propagation [40]. When light is incident on a material, part of it is reflected from the surface, part is transmitted through the medium, and part is absorbed. These processes are described by optical parameters such as reflectivity, absorption coefficient, refractive index, and extinction coefficient. The reflectivity $R(\omega)$ at normal incidence is calculated using

$$R(\omega) = \frac{[n(\omega) - 1]^2 + k^2(\omega)}{[n(\omega) + 1]^2 + k^2(\omega)}. \quad (8)$$

The absorption coefficient $\alpha(\omega)$, which describes how rapidly the intensity of light decreases inside the material, is given by

$$\alpha(\omega) = \frac{2\omega k(\omega)}{c}, \quad (9)$$

where c is the speed of light in vacuum. A high value of $\alpha(\omega)$ indicates strong absorption of photons, which is important for optoelectronic and photovoltaic applications. Therefore, the dielectric function, optical conductivity, refractive index, extinction coefficient, reflectivity, and absorption coefficient are strongly interrelated. Together, these parameters provide a complete description of the optical behaviour of a material, including how it absorbs, reflects, transmits, and dissipates electromagnetic radiation [41].

After calculating the optical response of the Nb_3X_8 and M_3YZ_7 layered kagome materials, we investigated their mechanical, piezoelectric, and flexoelectric responses using MD simulations

based on trained MTPs combined with the charge–dipole model. For uniaxial tensile deformation, a linear displacement field was applied along the y -direction according to $u_y = \varepsilon_{yy}y$, where u_y is the imposed displacement and ε_{yy} is the applied tensile strain. Figure 1 shows the schematic representation of Nb_3I_8 and Nb_3SeI_7 niobium-based kagome materials. For tensile stretching, we apply similar displacement of $0.0005 \text{ \AA}$ to the right edges of the atoms and $-0.0005 \text{ \AA}$ to the left edges of the atoms. Boundary atoms at the two opposite edges were displaced incrementally, while the interior atoms were allowed to relax at each loading step. For out-of-plane bending deformation, a quadratic displacement field was introduced as $u_z = \frac{1}{2}\kappa y^2$, where u_z is the out-of-plane displacement and κ is the bending curvature or strain-gradient parameter. The bending amplitude was selected to generate a measurable strain-gradient response while keeping the deformation within the linear flexoelectric regime. Because, it will help to increase the atomic forces and lead to bending deformation. After imposing the given displacement field to the atomic system, we fix the boundary atoms and allow the interior atoms to relax the energy fluctuations using the constant temperature simulations (NVT ensemble). Then we calculate the points charges and dipole moments for each atoms using the QP model. The steps for calculating the charges and dipole moments are described in the earlier studies [29, 42, 43] reference therein. This deformation produces a nonuniform strain field and enables the evaluation of the flexoelectric response. After each displacement step, the boundary atoms were kept fixed, while the remaining atoms were relaxed using NVT simulations. Atomic charges and dipole moments were then calculated using the QP model, and the total polarization was obtained from the total dipole moment normalized by the effective area of the relaxed atomic region.

The total polarization (P) of the configuration is determined via the following relation $P = \frac{1}{A} \sum_{i=1}^N (q_i r_i + p_i)$, where A represents the area and N is the total number of atoms inside the region (without considering the atoms near edges). In the present investigation, the generation of total polarization is the total dipole moment to the total area of the system. The constitutive relation for the polarization vector induced due to the flexo- and piezo-electricity effects may be written as:

$$P_i = e_{ijk} \varepsilon_{jk} + \mu_{ijkl} \frac{\partial \varepsilon_{jk}}{\partial r_l} \quad (10)$$

where e_{ijk} and μ_{ijkl} are the piezoelectric and flexoelectric tensors, respectively, ε_{jk} is the strain tensor, and $\partial \varepsilon_{jk} / \partial r_l$ is the strain-gradient tensor.

During bending deformation, the local atomic strain for each atom i can be calculated using the local deformation gradient F , and it depends on the initial and final coordinates of atoms. The local atomic strain tensor for atom i is as follows:

$$\varepsilon_i = \frac{1}{2} \left[F_i^T F_i - I \right] \quad (11)$$

where the term F_i describes the local deformation gradient, $\delta \varepsilon_{xz} / \delta x$, which provides the initial and final deformed coordinates of atoms, and the term I represent the identity matrix.

It must be important to note that the bending displacement of each inner atom due to bending deformations is in the principle curvature directions. The progressive bending displacement relations are already described in the previous studies and references [31, 42, 43, 44] therein. This contributes to an increase in atomic forces, which leads to bending deformation. In tensile and compressive loading circumstances, we provide displacement for ten-time steps and relaxation for 10,000 time steps. Data has been collected for each of the 1000 relaxation steps. Before applying

the loading conditions, atomic systems are equilibrated using the Nosé-Hoover thermostat and the velocity Verlet integration technique with a time increment of 0.5 fs. During relaxation, the edge atoms remain fixed [45]. After defining the displacement field in tensile and bending deformation, we fix the atomic systems near to edges and inner atoms to relax the energy fluctuations via the constant temperature simulation (NVT ensemble). We use a Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) [46] to perform the MD simulations.

Table 1: Comparison of lattice constants, vertical thickness parameter h , and total polarizability of non-Janus Nb_3X_8 and Janus M_3YZ_7 layered kagome monolayers obtained from DFT and the QP model [9, 47] .

S. No.	Materials	Lattice constant		Polarizability	
		a/b (Å)	h (Å)	$\alpha_{\text{total}}^{\text{DFT}}$ (Å ³)	$\alpha_{\text{total}}^{\text{CD}}$ (Å ³)
1	Nb_3Cl_8	6.74	3.44	40.14	40.05
2	Nb_3Br_8	7.11	3.72	55.19	54.61
3	Nb_3I_8	7.67	4.05	72.20	73.60
4	Nb_3SI_7	7.63	3.57	73.68	73.42
5	Nb_3SeI_7	7.66	3.73	74.61	73.89
6	Nb_3TeI_7	7.71	3.95	76.37	76.26
7	Nb_3SBr_7	7.17	3.44	61.60	61.22
8	Nb_3SeBr_7	7.20	3.60	62.72	62.43
9	Nb_3TeBr_7	7.26	3.82	64.57	64.22
10	Nb_3SCl_7	6.87	3.33	54.34	55.38
11	Nb_3SeCl_7	6.90	3.49	55.46	55.10
12	Nb_3TeCl_7	6.96	2.72	57.24	56.89
13	Ta_3SI_7	7.61	3.60	81.10	80.87
14	Ta_3SeI_7	7.63	3.77	80.46	79.54
15	Ta_3TeI_7	7.68	4.00	80.32	78.08
16	Ta_3SBr_7	7.15	3.47	64.10	63.53
17	Ta_3SeBr_7	7.18	3.64	65.26	64.76
18	Ta_3TeBr_7	7.23	3.87	67.55	66.58
19	Ta_3SCl_7	6.85	3.35	56.12	55.67
20	Ta_3SeCl_7	6.88	3.53	57.51	56.95
21	Ta_3TeCl_7	6.93	3.75	60.06	59.06

3. Results and discussion

In this work, DFT calculations and MD simulations were combined to investigate layered kagome niobium halides, namely non-Janus Nb_3X_8 and Janus chalcogenide-halides M_3YZ_7 . The DFT calculations were used to examine the phonon dispersion, optical response, and pressure- or volume-dependent thermodynamic properties. In contrast, MD simulations based on trained MTPs, coupled with the QP model, were employed to determine the mechanical, piezoelectric, and flexoelectric responses. The computational workflow was organized into four main stages.

First, phonon-dispersion calculations were performed to confirm the dynamical stability of the investigated materials and to validate the computational methodology. In addition, pressure/volume-dependent thermodynamic properties were evaluated using DFT under hydrostatic pressures ranging from 0 to 12 GPa, as described in the Supporting Information. Second, the optical properties of the non-Janus Nb_3X_8 and Janus M_3YZ_7 layered kagome materials were calculated from the frequency-dependent dielectric response. Third, uniaxial tensile deformation was applied to obtain the mechanical properties and piezoelectric response of each material. Finally, after validating the trained MTPs and the overall computational protocol, out-of-plane bending deformation was introduced to extract the flexoelectric coefficients after eliminating the contribution from piezoelectric polarization.

Optical anisotropy plays a crucial role in understanding the light-matter interaction of layered flat-band kagome materials and is directly related to their potential optoelectronic applications. The investigated optical parameters are the refractive index $n(\omega)$, extinction coefficient $k(\omega)$, absorption coefficient $\alpha(\omega)$, reflectivity $R(\omega)$, and the real and imaginary parts of the dielectric function, $\varepsilon_1(\omega)$ and $\varepsilon_2(\omega)$, respectively. In the present work, the optical properties of layered niobium kagome halides, Nb_3X_8 and chalcogenide-halides, M_3YZ_7 , are examined to clarify their anisotropic optical response (see Fig. 1). The optical properties of the layered kagome halides Nb_3X_8 were calculated for two different light polarization directions, namely in-plane ($E\parallel b$) and out-of-plane ($E\parallel c$), as shown in Fig. 2. The solid curves represent the optical response along the in-plane (b) direction, while the dashed curves represent the response along the out-of-plane (c) direction. The optical response of these layered kagome halides materials is described by the complex dielectric function, $\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega)$, where $\varepsilon_1(\omega)$ represents the dispersive part of the optical response and $\varepsilon_2(\omega)$ represents the absorptive part. The imaginary part originates mainly from interband electronic transitions between occupied valence-band states and unoccupied conduction-band states, while the real part can be obtained from the imaginary part through the Kramers–Kronig relation [48, 49]. From the dielectric function, other optical parameters such as the $n(\omega)$, $R(\omega)$, $k(\omega)$, and $\alpha(\omega)$ were derived (see Eqs. 1 to Eqs. 9). The directional dependence of the optical spectra originates from the layered structure and Nb-based kagome framework, where the electronic states near the band edges are mainly localized within the in-plane Nb network. Consequently, the in-plane optical response is generally stronger than the out-of-plane response. The optical behaviour is further tuned by chemical substitution: heavier and more polarizable atoms such as Te and I enhance the dielectric response, $n(\omega)$, $k(\omega)$, and $\alpha(\omega)$, whereas lighter atoms such as Cl and Br lead to a weaker response and a possible blue shift of optical features. These results indicate that Nb_3X_8 and M_3YZ_7 materials are promising anisotropic materials for polarization-sensitive photodetectors, optical modulators, birefringent components, and other layered optoelectronic devices.

3.1. Optical properties of Nb_3X_8

Figure 2 shows the calculated optical properties of layered kagome non-Janus Nb_3X_8 ($X = \text{Cl}, \text{Br}, \text{I}$) for light polarized along the in-plane ($E\parallel b$) and out-of-plane ($E\parallel c$) directions. A pronounced polarization anisotropy is observed for all optical quantities. In most of the investigated photon-energy range, the in-plane response is considerably stronger than the out-of-plane response. This anisotropy is a direct consequence of the layered van der Waals structure and the breathing-kagome Nb network, where the optically active states are mainly formed by in-plane Nb- d orbitals hybridized with halogen- p states [50, 41]. The weak out-of-plane response reflects the reduced orbital overlap and weaker electronic coupling along the c direction.

Figure 2(a) presents the refractive index $n(\omega)$, which measures the phase retardation of light inside the material. The in-plane refractive index is much larger and more dispersive than the out-of-plane component, indicating stronger electronic screening and slower light propagation within the kagome plane. Among the three compounds, Nb_3I_8 shows the largest refractive response, followed by Nb_3Br_8 and Nb_3Cl_8 . This trend is mainly associated with the increasing electronic deformability of the halogen sub-lattice from Cl to Br and I. The more spatially extended p -orbitals of heavier halogens respond more strongly to the oscillating electric field, thereby increasing the induced polarization and enhancing $n(\omega)$. The strong separation between the $E\parallel b$ and $E\parallel c$ components confirms the birefringent character of the Nb_3X_8 family, which is useful for polarization-sensitive optical devices.

The extinction coefficient $k(\omega)$, shown in Fig. 2(b), describes the attenuation of electromagnetic waves during propagation. Unlike $n(\omega)$, which reflects the dispersive part of the optical response, $k(\omega)$ is directly connected with optical loss. At low photon energies, $k(\omega)$ remains small because only weak interband transitions are available. As the photon energy increases, several intense peaks appear, indicating the activation of dipole-allowed transitions from occupied valence states to unoccupied conduction states. The stronger $E\parallel b$ peaks reveal that these optical losses are dominated by transitions within the Nb-based kagome layer. The broader and more intense response of Nb_3I_8 indicates that iodine not only enhances electronic polarizability but also increases the density and oscillator strength of optically active transitions in the visible and near-ultraviolet regions.

Figure 2(c) displays the absorption coefficient $\alpha(\omega)$. The absorption coefficient is on the order of 10^6 cm^{-1} , confirming strong light absorption in the visible and near-ultraviolet regions. For Nb_3Cl_8 , the main absorption features appear near 1.0, 2.4, 3.25, and 3.8 eV, whereas Nb_3Br_8 exhibits pronounced peaks around 1.1, 2.0, 3.0, and 3.5 eV. In Nb_3I_8 , dominant absorption peaks occur near 0.9 and 3.1 eV. The shift of the absorption edge and spectral peaks toward lower energy from Cl to Br and I indicates a progressive reduction in the effective transition energy between occupied and unoccupied states. This redshift is caused by the heavier halogen atoms, which introduce more diffuse p -states and stronger spin-orbit interaction, modifying the Nb- d /halogen- p hybridized bands. Therefore, Nb_3I_8 is more efficient for low-energy visible-light absorption, while Nb_3Cl_8 retains stronger features at relatively higher photon energies.

The reflectivity $R(\omega)$, shown in Fig. 2(d), also exhibits strong directional dependence. The in-plane reflectivity is larger than the out-of-plane reflectivity over most of the photon-energy range. This behaviour does not arise only from absorption; rather, it is controlled by the optical impedance mismatch between the material and vacuum. Because both $n(\omega)$ and $k(\omega)$ are larger for $E\parallel b$, the in-plane optical impedance differs more strongly from that of vacuum, leading to enhanced reflection. In the visible-energy region, the in-plane reflectivity is nearly twice the out-of-plane value. Among the three compounds, Nb_3I_8 generally shows the highest reflectivity, especially in energy regions where its extinction coefficient is large. This indicates stronger surface optical response and larger electromagnetic energy backscattering in the iodine-based compound.

Figures 2(e) and 2(f) show the real and imaginary parts of the dielectric function, $\epsilon_1(\omega)$ and $\epsilon_2(\omega)$, respectively. The real part, $\epsilon_1(\omega)$, describes the dispersive dielectric response and the ability of the material to store electromagnetic energy through polarization. The static dielectric constant $\epsilon_1(0)$ increases systematically from Nb_3Cl_8 to Nb_3I_8 . For the in-plane direction, $\epsilon_1(0)$ is approximately 16, 18, and 23 for Nb_3Cl_8 , Nb_3Br_8 , and Nb_3I_8 , respectively, while the corresponding out-of-plane values are approximately 7, 8, and 9.5. This large difference confirms strong dielectric anisotropy. The higher $\epsilon_1(0)$ of Nb_3I_8 indicates stronger low-frequency dielectric screen-

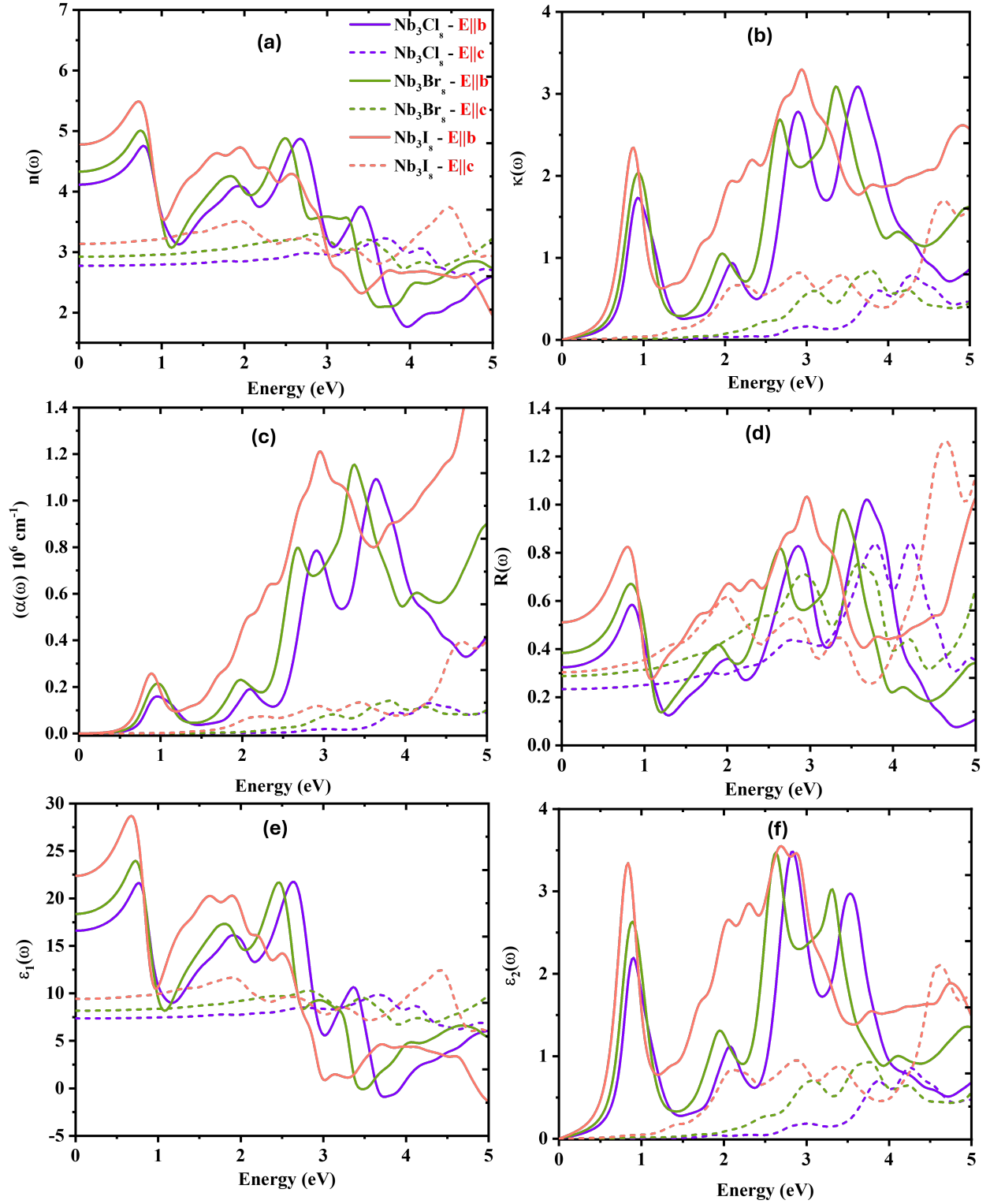


Fig. 2. Optical properties of non-Janus Nb_3X_8 ($\text{X} = \text{Cl}, \text{Br}, \text{I}$), where solid lines represent the response for $E||b$, while dashed lines represent the response for $E||c$. The panels show: (a) refractive index $n(\omega)$; (b) extinction coefficient $\kappa(\omega)$; (c) absorption coefficient $\alpha(\omega)$; (d) reflectivity $R(\omega)$; (e) real part of the dielectric function $\epsilon_1(\omega)$; and (f) imaginary part of the dielectric function $\epsilon_2(\omega)$.

ing, which is consistent with its larger refractive index. At higher photon energies, the oscillatory behaviour of $\varepsilon_1(\omega)$ originates from the dispersive response associated with interband transitions. In some regions, ε_1 decreases sharply and can approach small or negative values, suggesting strong screening and plasma-like optical behaviour. The imaginary dielectric component $\varepsilon_2(\omega)$, shown in Fig. 2(f), represents the absorptive part of the dielectric response. Peaks in $\varepsilon_2(\omega)$ identify photon energies where electronic transitions are most intense. The $E||b$ component exhibits much stronger and sharper peaks than the $E||c$ component, confirming that the dominant optical transitions occur within the in-plane kagome network. The strongest $\varepsilon_2(\omega)$ response is found for Nb_3I_8 , followed by Nb_3Br_8 and Nb_3Cl_8 . This trend reflects the increasing transition strength and enhanced spin–orbit-assisted band mixing in heavier-halogen systems. Thus, while $\varepsilon_1(\omega)$ highlights the composition-dependent dielectric screening, $\varepsilon_2(\omega)$ directly reveals the enhanced interband absorption channels.

Overall, Fig. 2 demonstrates that Nb_3X_8 compounds exhibit strong anisotropic optical behaviour. The optical response is consistently stronger for $E||b$ than for $E||c$, confirming that light–matter interaction is mainly governed by the in-plane Nb-based breathing-kagome framework. The overall optical strength follows the order $\text{Nb}_3\text{I}_8 > \text{Nb}_3\text{Br}_8 > \text{Nb}_3\text{Cl}_8$, but each optical quantity reflects a different physical aspect: $n(\omega)$ is enhanced by dielectric screening, $k(\omega)$ by optical attenuation and transition probability, $\alpha(\omega)$ by the absorption edge and interband transition energy, $R(\omega)$ by optical impedance mismatch, $\varepsilon_1(\omega)$ by dispersive polarization, and $\varepsilon_2(\omega)$ by the strength of interband absorption. The calculated optical response of Nb_3Cl_8 shows excellent consistency with the previous first-principles study of Bouhmouche *et al.* [17], which reported strong optical anisotropy with a much larger in-plane response than the out-of-plane component. In agreement with that work, our Nb_3Cl_8 spectra show a larger in-plane refractive index, dielectric response, reflectivity, and absorption coefficient, confirming that the light–matter interaction is dominated by the Nb-based kagome plane. The reported in-plane/out-of-plane contrast in $n(\omega)$, optical conductivity, and absorption coefficient for Nb_3Cl_8 provides a direct benchmark for validating our optical calculations [17]. After reproducing the key anisotropic features of Nb_3Cl_8 , we extend the same computational framework to Nb_3Br_8 and Nb_3I_8 , revealing a systematic enhancement of the optical response with heavier halogen substitution. The large absorption coefficient, strong birefringence, and pronounced dielectric anisotropy suggest that Nb_3X_8 compounds are promising candidates for visible-light photodetectors, anisotropic optoelectronic devices, polarization-sensitive optical sensors, and light-harvesting applications [8, 15, 51].

3.2. Optical properties of Nb_3YZ_7

Next, we examine the optical response of Janus layered kagome Nb_3YZ_7 compounds, represented here by Nb_3SeI_7 , Nb_3SeBr_7 , and Nb_3TeI_7 , as shown in Fig. 3. The spectra are resolved for light polarized along the in-plane ($E||b$) and out-of-plane ($E||c$) directions in order to clarify the anisotropic light–matter interaction. For all three compounds, the in-plane optical response is stronger than the out-of-plane response over most of the photon-energy range. This anisotropy originates from the layered crystal structure and the Nb-based breathing-kagome framework, where the optically active states are mainly distributed within the kagome plane. In contrast, the out-of-plane response is weaker because the orbital overlap along the c -direction is reduced and the electronic coupling between the two surfaces is comparatively weaker. The Janus composition further introduces asymmetric chalcogen–halogen environments, which modifies the local electrostatic potential and changes the optical transition strength.

Figure 3(a) shows the refractive index $n(\omega)$, which describes the phase velocity of light inside the material. The in-plane refractive index is larger and more dispersive than the out-of-plane component, indicating stronger dielectric screening and larger phase retardation for light polarized within the kagome plane. Among the three compounds, Nb_3SeI_7 and Nb_3TeI_7 show stronger refractive-index features than Nb_3SeBr_7 . This behaviour reflects the larger electronic deformability of I-containing systems compared with the Br-containing compound. In addition, the Te- p states in Nb_3TeI_7 are more spatially extended than Se- p states, which increases dielectric screening and strengthens the refractive response in selected energy regions. Thus, the refractive-index anisotropy is mainly governed by direction-dependent electronic polarizability and orbital delocalization within the Janus kagome layer.

Figure 3(b) presents the extinction coefficient $k(\omega)$, which measures optical attenuation during propagation. In contrast to $n(\omega)$, which reflects the dispersive part of the optical response, $k(\omega)$ is associated with optical loss and the probability of interband electronic excitation. The small values at low photon energies indicate weak attenuation below the main optical-transition region. With increasing photon energy, intense peaks appear because electrons are excited from occupied valence-band states to unoccupied conduction-band states. The strongest peaks occur for the in-plane polarization, confirming that the dominant absorption channels are located within the Nb-based kagome network. Nb_3SeI_7 exhibits the strongest extinction peak, whereas Nb_3TeI_7 shows a broad response over a wider energy range. Nb_3SeBr_7 displays a comparatively weaker extinction coefficient because Br is less polarizable than I and produces weaker dipole-allowed transition strength.

The absorption coefficient $\alpha(\omega)$, shown in Fig. 3(c), describes the rate at which light intensity decays inside the material. Although $\alpha(\omega)$ is related to $k(\omega)$, it also contains the photon-frequency factor and therefore emphasizes higher-energy transitions more strongly. The calculated $\alpha(\omega)$ values reach the order of 10^6 cm^{-1} , indicating strong absorption in the visible and near-ultraviolet regions. The absorption is much stronger for $E||b$ than for $E||c$, demonstrating that these Janus kagome compounds are more efficient absorbers for in-plane-polarized light. Among the three systems, Nb_3SeI_7 shows the largest absorption peak, while Nb_3TeI_7 provides a comparable and broader absorption window. The Br-containing Nb_3SeBr_7 compound exhibits weaker absorption, consistent with its reduced polarizability and weaker optical transition strength. The difference between I- and Br-containing systems indicates that halogen substitution is an effective route to tune the absorption edge and the intensity of visible-light absorption.

Figure 3(d) shows the reflectivity $R(\omega)$. The reflectivity is determined by the optical impedance mismatch between the material and vacuum and therefore depends jointly on $n(\omega)$ and $k(\omega)$. The in-plane reflectivity is larger than the out-of-plane reflectivity in most energy regions because the in-plane components have larger refractive index and stronger attenuation. This indicates that light polarized along the kagome plane experiences stronger surface backscattering. Nb_3SeI_7 generally exhibits the highest reflectivity in the main optical-transition region, while Nb_3TeI_7 remains competitive over a broader energy range. The lower reflectivity of Nb_3SeBr_7 is consistent with its smaller extinction coefficient and weaker dielectric screening. Therefore, the reflectivity trend is controlled not only by composition-dependent polarizability but also by the surface optical impedance created by the anisotropic dielectric response.

Figures 3(e) and 3(f) display the real and imaginary parts of the dielectric function, $\epsilon_1(\omega)$ and $\epsilon_2(\omega)$, respectively. The real component $\epsilon_1(\omega)$ represents the dispersive dielectric response and the ability of the material to store electromagnetic energy through induced polarization. The

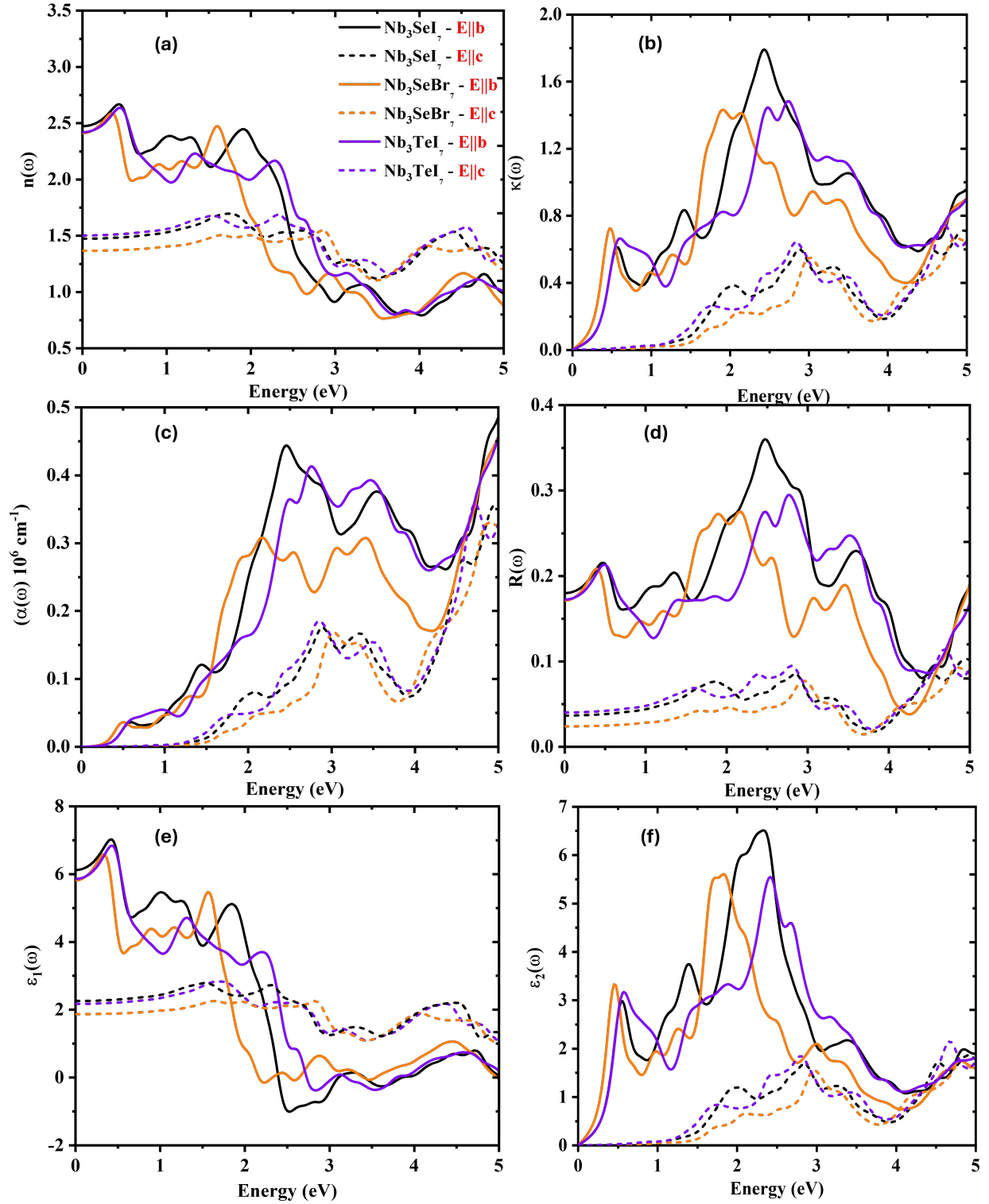


Fig. 3. Optical properties of Janus Nb_3YZ_7 compounds Nb_3SeI_7 , Nb_3SeBr_7 , and Nb_3TeI_7 , where solid lines represent the response for $E||b$, while dashed lines represent the response for $E||c$. The panels show: (a) refractive index $n(\omega)$; (b) extinction coefficient $k(\omega)$; (c) absorption coefficient $\alpha(\omega)$; (d) reflectivity $R(\omega)$; (e) real part of the dielectric function $\epsilon_1(\omega)$; and (f) imaginary part of the dielectric function $\epsilon_2(\omega)$.

static dielectric response is clearly anisotropic: the in-plane $\epsilon_1(0)$ values are much larger than the corresponding out-of-plane values. This confirms that low-frequency dielectric screening is dominated by the in-plane kagome framework. The oscillations and sharp decreases in $\epsilon_1(\omega)$ at higher photon energies arise from the Kramers–Kronig-related dispersive response of strong interband absorption peaks. In some energy windows, $\epsilon_1(\omega)$ approaches small or negative values, indicating strong screening and plasma-like optical behaviour for in-plane-polarized light.

The imaginary dielectric component $\epsilon_2(\omega)$ represents the absorptive part of the dielectric response and directly identifies the energies at which interband transitions are strongest. The in-plane $\epsilon_2(\omega)$ spectra show much larger peaks than the out-of-plane spectra, confirming that optical absorption mainly arises from transitions within the Nb-cluster kagome layer. Nb_3SeI_7 shows the strongest $\epsilon_2(\omega)$ peak, indicating a large joint density of states and strong optical transition matrix elements in the visible-energy region. Nb_3TeI_7 shows a broad and intense response due to the participation of diffuse Te- p orbitals and stronger spin–orbit-induced band mixing. In contrast, Nb_3SeBr_7 exhibits a weaker and more shifted dielectric-loss spectrum because the Br states are less polarizable and couple less strongly with the Nb- d states.

Overall, Fig. 3 demonstrates that Janus Nb_3YZ_7 compounds possess strong optical anisotropy and composition-dependent optical tunability. The optical response is consistently stronger for $E||b$ than for $E||c$, confirming that light–matter interaction is dominated by the in-plane breathing-kagome Nb network. However, the physical origin of each optical quantity is different: $n(\omega)$ reflects phase retardation and dielectric screening; $k(\omega)$ represents optical attenuation and transition probability; $\alpha(\omega)$ is governed by absorption onset, photon energy, and interband transition strength; $R(\omega)$ is controlled by optical impedance mismatch; $\epsilon_1(\omega)$ describes dispersive polarization and screening; and $\epsilon_2(\omega)$ measures the strength of interband absorption. Among the studied compounds, Nb_3SeI_7 exhibits the strongest peak optical response, Nb_3TeI_7 provides a broader absorption and dielectric window, and Nb_3SeBr_7 shows a comparatively weaker but still strongly anisotropic response. These results suggest that Janus Nb_3YZ_7 monolayers are promising candidates for anisotropic photodetectors, polarization-sensitive optical devices, nonlinear optics, and light-harvesting applications [22, 52, 53].

3.3. Electromechanical response of Nb_3X_8 and M_3YZ_7

Finally, we investigated the electromechanical response of the layered kagome non-Janus Nb_3X_8 and Janus M_3YZ_7 materials by applying uniaxial tensile strain along the y -direction. The corresponding stress–strain curves for representative systems are shown in Fig. 4(a) within the elastic strain range up to $\epsilon = 0.03$, while Fig. 4(c) summarizes the Young’s modulus (E_{yy} , GPa) obtained from trained MTP for the full material set with QP model. For the conversion of stress into the standard three-dimensional unit of GPa, the real volume of the deformed monolayer was considered instead of the total simulation-cell volume, which contains an artificial vacuum region. The real in-plane area of each deformed nanosheet was calculated from the simulation box dimensions along the in-plane directions (y -direction). The effective volume at each strain was then obtained by multiplying this area by the effective monolayer thickness. Following the approach used for 2D layered systems, the effective thickness was estimated from the normal distance between the outermost boundary atoms together with their effective vdW diameter [9]. The resulting effective thicknesses of the stress-free Nb_3X_8 and M_3YZ_7 monolayers are listed in Table 1.

In this work, the vdW-corrected monolayer thickness was calculated as $t_{\text{vdW}} = h + r_{\text{vdW}}^{\text{top}} + r_{\text{vdW}}^{\text{bottom}}$, where h is the vertical distance between the outermost atomic planes of the relaxed monolayer.

For non-Janus Nb_3X_8 monolayers, $t_{\text{vdW}} = h + 2r_{\text{vdW}}^{\text{X}}$, whereas for Janus M_3YZ_7 monolayers, $t_{\text{vdW}} = h + r_{\text{vdW}}^{\text{Y}} + r_{\text{vdW}}^{\text{Z}}$. The vdW radii used for S, Se, Te, Cl, Br, and I were 1.80, 1.90, 2.06, 1.75, 1.85, and 1.98, respectively. As shown in Fig. 4(a), the stress increases almost linearly with applied strain for all selected compounds, confirming that the systems remain within the elastic regime over the considered strain range. The slope of each stress–strain curve represents the Young’s modulus. Among the representative materials, Nb_3Cl_8 exhibits the steepest slope, indicating the largest Young’s modulus and therefore the strongest in-plane stiffness. This higher stiffness can be attributed to the relatively light Cl atoms, shorter Nb–Cl bond lengths, and stronger effective bond force constants [9]. In contrast, the chalcogenide–halide and Ta-based compounds display smaller slopes, revealing a softer mechanical response. This softening is mainly associated with larger atomic radii, longer metal–halogen or metal–chalcogen bonds, weaker effective bonding interactions, and enhanced lattice flexibility [10]. The trend observed in Fig. 4(a) follows the expected mechanical behaviour of layered 2D materials. Replacing lighter atoms with heavier and more polarizable atoms generally reduces the effective force constants and increases the bond lengths, leading to lattice softening. Consequently, Cl-containing systems are mechanically stiffer than the corresponding Br- and I-containing compounds. Similarly, Janus chalcogenide–halides exhibit greater structural flexibility because their asymmetric bonding environment promotes nonuniform internal relaxation under strain. This behaviour is consistent with previous reports on M_3QX_7 monolayers, whose 2D Young’s moduli are considerably smaller than those of highly stiff 2D materials such as graphene and are also lower than many TMDs [2, 10].

Figure 4(c) compares the Young’s modulus obtained from trained MTP calculations with the corresponding DFT values [47]. The close agreement between the two methods demonstrates that the trained MTP accurately reproduces the first-principles mechanical stiffness. In most cases, the MTP values are slightly lower than the DFT values, which is reasonable because MLIPs can slightly smooth the potential-energy surface. Nevertheless, the relative ordering of the materials is well preserved, confirming the reliability of the trained potential for describing elastic deformation. For the non-Janus halides, the Young’s modulus of Nb_3Cl_8 is approximately 1.10 times higher than that of Nb_3Br_8 and 1.27 times higher than that of Nb_3I_8 , clearly reflecting progressive lattice softening from Cl to Br and then to I. This trend originates from the increasing halogen atomic size, longer Nb–X bond lengths, and decreasing effective bond strength. For the Janus chalcogenide–halides M_3YZ_7 , the stiffness is governed by the combined effect of transition-metal species, chalcogen/halogen chemistry, and bonding asymmetry. For example, the MTP-calculated Young’s modulus of Nb_3SCl_7 is approximately 1.36 times larger than that of Ta_3SCl_7 , indicating that the Nb-based compound possesses stronger in-plane stiffness. This higher stiffness reflects the stronger effective Nb–S/Nb–Cl bonding network, shorter bond lengths, and larger in-plane force constants, whereas the Ta-based analogue exhibits enhanced lattice softness due to weaker effective Ta–chalcogen/halogen interactions and the larger atomic mass of Ta.

Furthermore, we evaluated the piezoelectric coefficient e_{yyy} for the non-Janus and Janus layered kagome compounds from the linear slope of the polarization–strain curves, as shown in Fig. 4(b). The nearly linear variation of polarization with strain confirms that the investigated systems remain in the linear piezoelectric regime over the considered strain range. As shown in Fig. 4(b), the Janus compounds exhibit a stronger polarization response than the non-Janus systems. Among the selected materials, Nb_3SeI_7 displays one of the largest strain-induced polarization changes, indicating a strong piezoelectric response. This enhancement originates from its Janus-type asymmetric structure, in which chemically different atomic layers on the two sides of the monolayer break out-

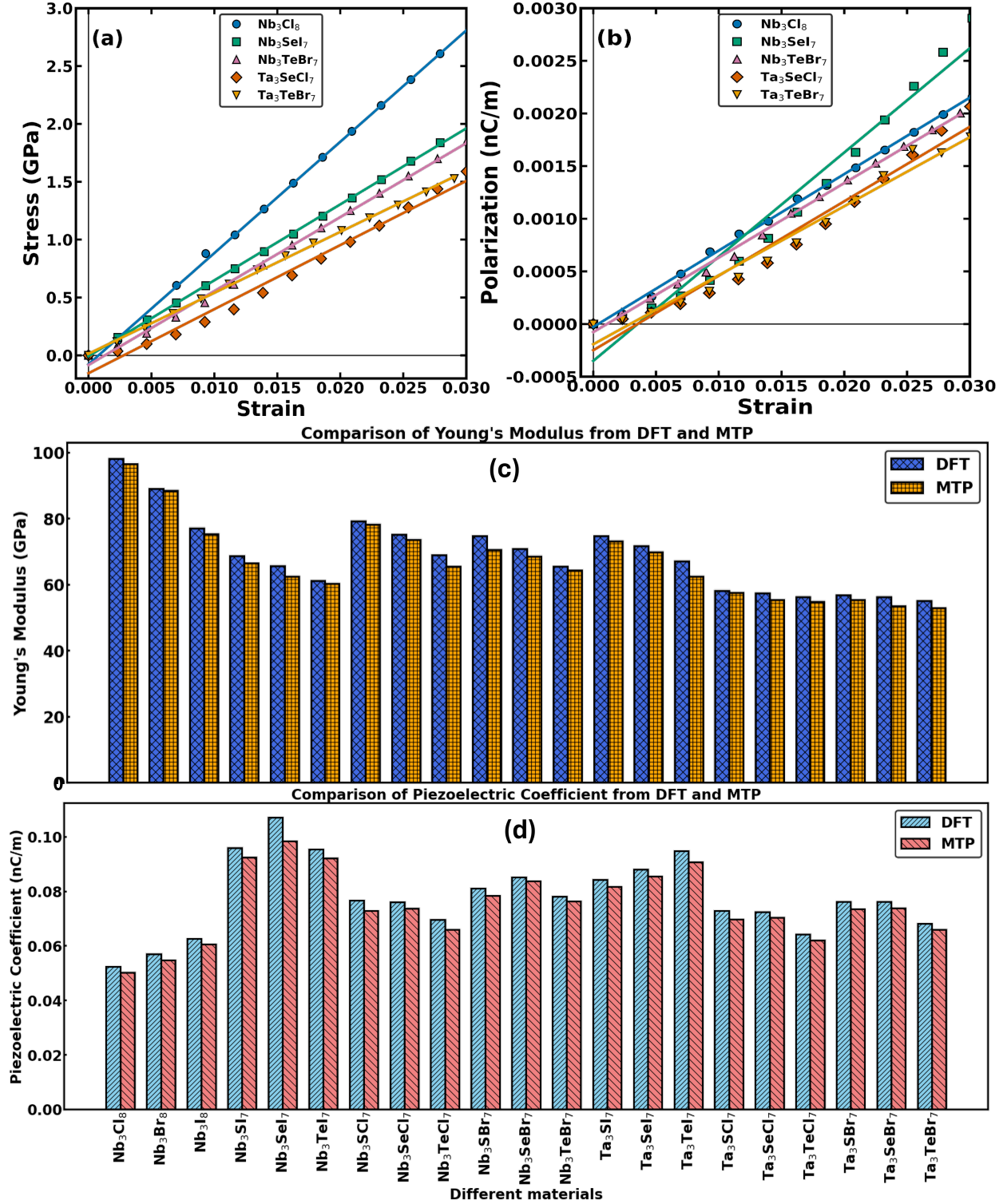


Fig. 4. Mechanical and piezoelectric responses of layered kagome halides and chalcogenide-halides. (a) Stress–strain curves for selected compounds, where the slope gives the Young’s modulus. (b) Polarization as a function of strain, where the slope corresponds to the piezoelectric coefficient. (c) Comparison of Young’s modulus obtained from DFT and MTP calculations for the investigated materials. (d) Comparison of the piezoelectric coefficient obtained from DFT and MTP calculations. The close agreement between DFT and MTP confirms the reliability of the trained potential, while the enhanced piezoelectric response of Janus compounds originates from broken mirror symmetry and strain-induced polarization [9, 47].

of-plane mirror symmetry and produce an intrinsic polar environment [52, 54]. The coexistence of Se and I atoms creates an asymmetric charge distribution and enhances the internal electric-field variation under strain, thereby increasing the strain-induced polarization. To further understand the origin of the piezoelectric response, we analyzed the atomic polarizability and the strain-induced change in the local electric field, ΔE_y , obtained from the QP model. The details of this methodology have been described in previous studies [55, 42]. As listed in Table 1, the polarizabilities obtained from DFT are in good agreement with those calculated using the QP model, confirming the reliability of the adopted parameterization. For the Janus Nb_3YZ_7 compounds, the unit-cell polarizabilities of Nb_3SeI_7 , Nb_3SeBr_7 , and Nb_3SeCl_7 are $74.61\text{\AA}^3$, $62.72\text{\AA}^3$, and $55.46\text{\AA}^3$, respectively. Over the strain range from $\varepsilon = 0$ to 0.025, the corresponding changes in the local electric field, ΔE_y , are $162.45\text{ V/\AA}$, $145.12\text{ V/\AA}$, and $151.49\text{ V/\AA}$, respectively. These results indicate that the piezoelectric polarization is governed by a combined effect of polarizability and strain-induced electric-field variation. A larger ΔE_y tends to enhance the polarization response, whereas a larger polarizability can partially screen the local field and reduce the effective field-induced polarization change. Therefore, the polarization response cannot be interpreted from polarizability or ΔE_y alone; rather, it results from the balance between local-field enhancement, charge redistribution, and dielectric screening. For example, the relatively smaller ΔE_y of Nb_3SeBr_7 reduces its polarization response compared with systems having stronger local-field variation. In contrast, although Nb_3SeCl_7 has a smaller polarizability, its relatively large ΔE_y supports a stronger strain-induced polarization. Thus, the piezoelectric response is enhanced by strong strain-induced local electric fields and reduced dielectric screening. From the slope of the polarization–strain curves, the piezoelectric coefficients of Nb_3SeBr_7 and Nb_3SeCl_7 are obtained as 0.076 and 0.085 nC/m, respectively, as shown in Fig. 4(d). These values are approximately 1.40 and 1.25 times lower than that of Nb_3SeI_7 , respectively. This trend demonstrates that, although Nb_3SeI_7 possesses a large intrinsic polarizability and strong Janus asymmetry, the final piezoelectric coefficient is controlled by the interplay between atomic polarizability, local electric-field variation, and strain-induced internal structural relaxation.

For the non-Janus Nb_3X_8 ($X = \text{Cl}, \text{Br}, \text{I}$) compounds, the polarization response originates from strain-induced charge redistribution between Nb and halogen atoms. Because Nb is less electronegative than the halogens, electron density is transferred from Nb toward X, producing local polar Nb–X bonds. However, the similar halogen environments on the two sides of the Nb layer lead to partial cancellation of opposite local dipoles, which strongly reduces the net polarization. As the halogen changes from Cl to Br and I, the lattice parameter increases and the lattice becomes softer because of larger atomic radii and longer Nb–X bonds. This softness allows larger atomic displacements under strain, but the weaker charge transfer and dipole cancellation limit the total piezoelectric response. It can also be seen from Fig. 4(b) that the non-Janus Nb_3X_8 systems show comparatively smaller polarization responses than the Janus chalcogenide-halides. The increase in piezoelectric response from non-Janus halides to Janus M_3YZ_7 monolayers can be understood from the combined effect of structural polarity and mechanical softness. A soft lattice can undergo larger internal atomic displacement under small strain, while broken mirror symmetry prevents cancellation of the induced dipoles and allows a larger net polarization. Therefore, although the lattice parameter increases and the lattice softens from Cl to Br and I in the non-Janus systems, their weaker structural asymmetry results in smaller polarization–strain slopes compared with Janus chalcogenide-halides. Therefore, materials with larger surface asymmetry, stronger charge redistribution, and smaller elastic stiffness tend to show enhanced piezoelectric strain response. This

explains why Janus compounds such as M_3YZ_7 monolayers exhibit larger piezoelectric responses than centrosymmetric or weakly polar systems. Figure 4(d) compares the piezoelectric coefficients obtained from DFT and MTP calculations. Similar to the Young's modulus comparison, the MTP results closely follow the DFT values, confirming that the trained potential combined with the QP model can reproduce the electromechanical response with good accuracy. The largest piezoelectric coefficients are obtained for polar Janus structures, especially those containing Se or Te together with Br or I. These compounds possess stronger inversion- and mirror-symmetry breaking, which enhances strain-induced polarization. The smaller piezoelectric coefficients in some Ta-based or Cl-containing systems can be attributed to weaker polarization changes under strain and partial cancellation between electronic and ionic contributions. From the slope of the polarization-strain curves, the piezoelectric coefficients of Nb_3SeBr_7 and Nb_3SeCl_7 are obtained as 0.076 and 0.085 nC/m, respectively. Compared with these values, Nb_3SeI_7 exhibits a larger piezoelectric coefficient, being approximately 1.40 and 1.26 times higher than those of Nb_3SeBr_7 and Nb_3SeCl_7 , respectively. This trend demonstrates that the piezoelectric response is not governed by atomic polarizability alone, but by the combined influence of Janus structural polarity, strain-induced local electric-field variation, charge redistribution, dielectric screening, and internal atomic relaxation [44, 42].

Having established the reliability of the MTP-QP framework for describing elastic deformation and strain-induced polarization, we next evaluated the flexoelectric response of non-Janus Nb_3X_8 and Janus M_3YZ_7 layered kagome monolayers under controlled bending-induced strain gradients. The objective was to evaluate the coupling between the strain gradient (nm^{-1}), and the induced in-plane and out-of-plane polarization components under bending deformation, as illustrated in Fig. 6. Unlike piezoelectricity, which requires broken inversion symmetry, flexoelectricity is allowed in materials of any symmetry because it is induced by a non-uniform strain field [29, 30]. Therefore, both Janus and non-Janus layered kagome systems can exhibit flexoelectric polarization when subjected to bending deformation (see Fig. 6). To generate bending deformation, the monolayer was first compressed in the in-plane direction until buckling occurred at a critical compressive strain. Beyond this point, the structure naturally adopted a bent configuration, following the methodology reported in previous studies [42, 43]. The resulting non-uniform deformation produces a spatially varying strain field and breaks inversion symmetry locally, thereby inducing an out-of-plane polarization component P_z . In this geometry, the dominant strain component generated by the out-of-plane displacement field u_z is ϵ_{yz} . At the same time, the antisymmetric strain distribution across the structure leads to cancellation of the net uniform strain contribution, which minimizes the piezoelectric component of the polarization. This analysis confirms that the bending deformation produces a controlled strain gradient rather than a dominant uniform tensile or compressive strain. Therefore, the induced P_z can be attributed primarily to the flexoelectric response. The initial and bent configurations of the selected non-Janus Nb_3X_8 and Janus M_3YZ_7 layered kagome compounds are shown in Fig. 6.

Figure 5(a) shows the variation of P_z with strain gradient for representative kagome monolayers. The nearly linear dependence of P_z on strain gradient confirms that the selected deformation range remains within the linear flexoelectric regime. The slope of each fitted line gives the corresponding flexoelectric coefficient (μ_{yz}). Among the representative systems, Nb_3TeBr_7 exhibits the steepest slope, indicating the largest bending-induced polarization response. This enhancement is mainly associated with the combined effect of Janus structural asymmetry, large contribution of electric fields, and lattice softness. In contrast, Nb_3SeI_7 and Ta_3SeCl_7 show smaller slopes, sug-

gesting weaker strain-gradient-induced polarization under the same bending condition. According to QP model, we obtained the electric field dependence (E_y^q and E_y^p) factors on the generation of the dipole moment p_z along the z-direction of the systems. For a strain gradient of $k=0.030$ 1/nm, the total electric field difference ΔE_z between initial to deformed configurations in Nb_3TeBr_7 is 125.39 V/Å which is higher than Ta_3SeCl_7 (88.37 V/Å) and Ta_3TeBr_7 (103.24 V/Å). This contribution of high electric field in Nb_3TeBr_7 significantly enhances the polarizability which develops a change in the dipole moment Δp_z as 261 eÅ, which is 81692 times higher than Ta_3TeBr_7 and 5647 times higher than Ta_3SeCl_7 monolayers. From Fig. 5(b), we estimate the different flexoelectric coefficients μ_{yzyz} and μ_{zzyy} for Janus and non-Janus kagome monolayers. A similar conclusion is obtained when comparing the chalcogen dependence for a fixed transition-metal and halogen species. As shown in Table 1, the flexoelectric response is more complex than a simple polarizability-driven trend. Although heavier chalcogen atoms such as Te possess larger atomic polarizability, their larger ionic radius and stronger dielectric screening can partially reduce the effective local electric field generated under bending. In contrast, lighter chalcogen atoms such as S can form shorter and stronger metal–chalcogen bonds, producing larger bond-force constants and more efficient strain-gradient-induced charge redistribution. Therefore, the flexoelectric coefficient is governed by the balance among bond stiffness, Janus asymmetry, charge transfer, dielectric screening, and curvature-induced internal relaxation. A comparison between Nb- and Ta-based Janus compounds further shows that Nb-based systems generally exhibit larger flexoelectric coefficients (see Fig. 5(b)). This trend can be attributed to the stronger effective Nb–chalcogen/halogen bonding network, which produces larger local-field variation and stronger curvature-induced charge redistribution during bending. In contrast, the Ta-based analogues remain flexoelectrically active but show relatively smaller responses, indicating that the Ta-centered bonding environment leads to weaker strain-gradient-induced polarization. Overall, Fig. 5 demonstrates that Janus polarity, bonding asymmetry, lattice softness, and local dielectric screening are the key microscopic factors controlling flexoelectricity in layered kagome monolayers.

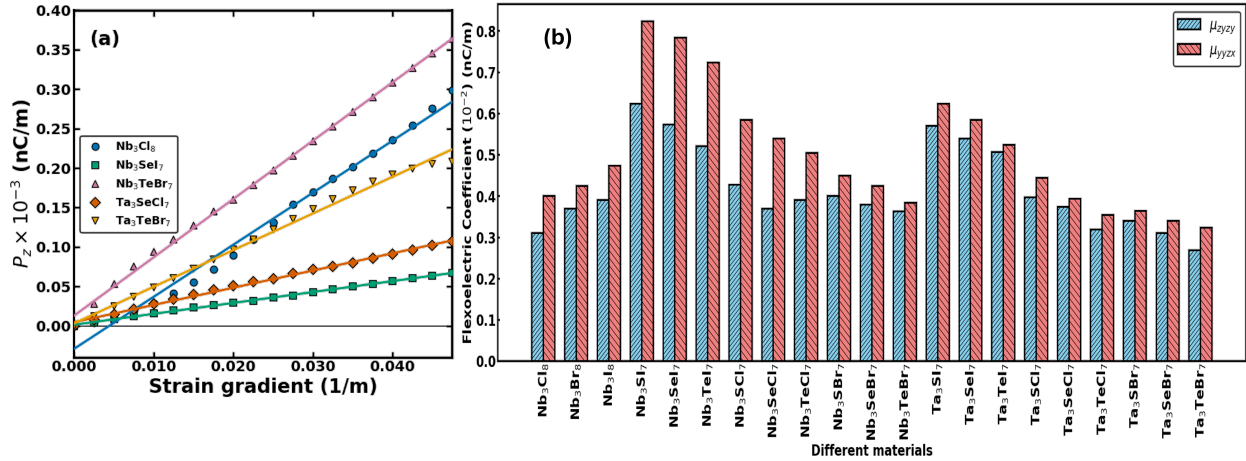


Fig. 5. Flexoelectric response of layered kagome halides and chalcogenide-halides. (a) Induced out-of-plane polarization P_z as a function of strain gradient κ for representative monolayers. The linear fits confirm the linear flexoelectric regime, and the slopes provide the flexoelectric coefficients. (b) Comparison of the calculated flexoelectric tensor components for non-Janus Nb_3X_8 and Janus M_3YZ_7 monolayers. The larger response of Janus I-containing compounds originates from enhanced structural polarity, large atomic polarizability, and curvature-induced charge redistribution.

The present electromechanical trends are consistent with recent experimental and theoretical reports on Janus kagome and Janus 2D materials. Wang *et al.* experimentally demonstrated robust out-of-plane piezoelectricity in Janus Nb₃SeI₇, with $|d_{33}^{\text{eff}}| = 0.76 \pm 0.11$ pm/V, confirming that broken out-of-plane mirror symmetry and non-canceling Nb–Se/Nb–I bonding generate strong vertical electromechanical coupling. Although d_{33} cannot be directly converted into an intrinsic flexoelectric coefficient, it provides experimental evidence that Janus Nb₃SeI₇ has a strong polarity-driven electromechanical response. Therefore, the enhanced flexoelectric response obtained here for bent Janus M₃YZ₇ monolayers is physically consistent with the experimentally observed vertical piezoelectric activity of Nb₃SeI₇. This comparison supports the role of Janus asymmetry, local charge redistribution, and lattice flexibility in amplifying strain- and strain-gradient-induced polarization. Similarly, inversion-asymmetric M₃QX₇ kagome monolayers have been predicted to host Rashba splitting and 2D piezoelectricity, demonstrating that structural polarity is a key design principle in this materials family [56]. In broader Janus 2D systems, Janus TMDs and Janus CrOFBr monolayers also show enhanced out-of-plane piezoelectric responses due to broken mirror symmetry and asymmetric charge redistribution [57, 58]. For flexoelectricity, previous charge–dipole simulations showed that bending-induced polarization is strongly enhanced in TMDs compared with graphene and becomes even larger in Janus TMDs because structural asymmetry strengthens σ – σ and π – σ electromechanical coupling [31, 59]. Therefore, the enhanced flexoelectric coefficients obtained here for Janus M₃YZ₇ monolayers are physically consistent with the established role of broken out-of-plane symmetry, charge redistribution, and lattice softness in amplifying strain- and strain-gradient-induced polarization.

Figure 5(b) compares the calculated flexoelectric coefficients of all investigated Nb₃X₈ and M₃YZ₇ monolayers. For each material, the μ_{zyy} -type component is larger than the corresponding μ_{zzy} -type component, indicating that the polarization response is more sensitive to shear-like bending strain gradients than to the normal bending component. This anisotropy originates from the low-dimensional layered geometry, where bending mainly redistributes charge through out-of-plane atomic displacements coupled with in-plane bond distortion. Therefore, the flexoelectric response is not isotropic, but strongly depends on the direction of the strain-gradient tensor and the local bonding environment. For the non-Janus Nb₃X₈ compounds, the flexoelectric coefficient increases from Nb₃Cl₈ to Nb₃I₈. This trend follows the progressive increase in halogen atomic radius and polarizability. Heavier halogens possess more diffuse *p*-orbitals, which enhance strain-gradient-induced charge redistribution. At the same time, the longer Nb–X bonds in Br- and I-containing systems make the lattice more deformable under bending, allowing larger internal atomic displacement. Consequently, Nb₃I₈ shows a larger flexoelectric response than Nb₃Cl₈. The Janus M₃YZ₇ monolayers exhibit a stronger and more tunable flexoelectric response than the non-Janus halides. The largest coefficients are observed for I-containing Nb-based Janus systems, particularly Nb₃SI₇, Nb₃SeI₇, and Nb₃TeI₇. Their enhanced response can be attributed to the built-in out-of-plane polarity created by chemically different chalcogen and halogen surfaces. Under bending, this intrinsic asymmetry prevents complete cancellation of local dipoles and promotes a larger net P_z . In addition, the large polarizability of iodine strengthens local electric-field variation, while the soft metal–chalcogen/halogen bonding network enables larger curvature-induced structural relaxation. These effects act cooperatively to increase μ . The magnitude of the present flexoelectric coefficients is comparable to those reported for several representative 2D materials. Previous atomistic QP simulations showed that graphene has a relatively weak flexoelectric response because of weak π – σ coupling, whereas transition-metal dichalcogenides such

as MoS₂ exhibit larger flexoelectric coefficients because bending enhances charge transfer and structural asymmetry [31]. First-principles studies also demonstrated that wrinkling and corrugation can generate sizeable out-of-plane flexoelectric polarization in monolayer TMDs [60], while h-BN monolayers show a stronger flexoelectric response than graphene due to their polar B–N bonds [61, 62]. Moreover, Janus TMDs were reported to possess substantially enhanced flexoelectric constants because their intrinsic vertical asymmetry amplifies σ – σ and π – σ electromechanical coupling [59]. The present kagome M₃YZ₇ monolayers follow the same physical principle: Janus asymmetry, large atomic polarizability, and soft lattice deformation cooperate to produce enhanced strain-gradient-induced polarization. This places layered kagome niobium halides and chalcogenide-halides among promising 2D flexoelectric materials for curvature-sensitive sensors, bending-driven nanogenerators, and flexible electromechanical devices.

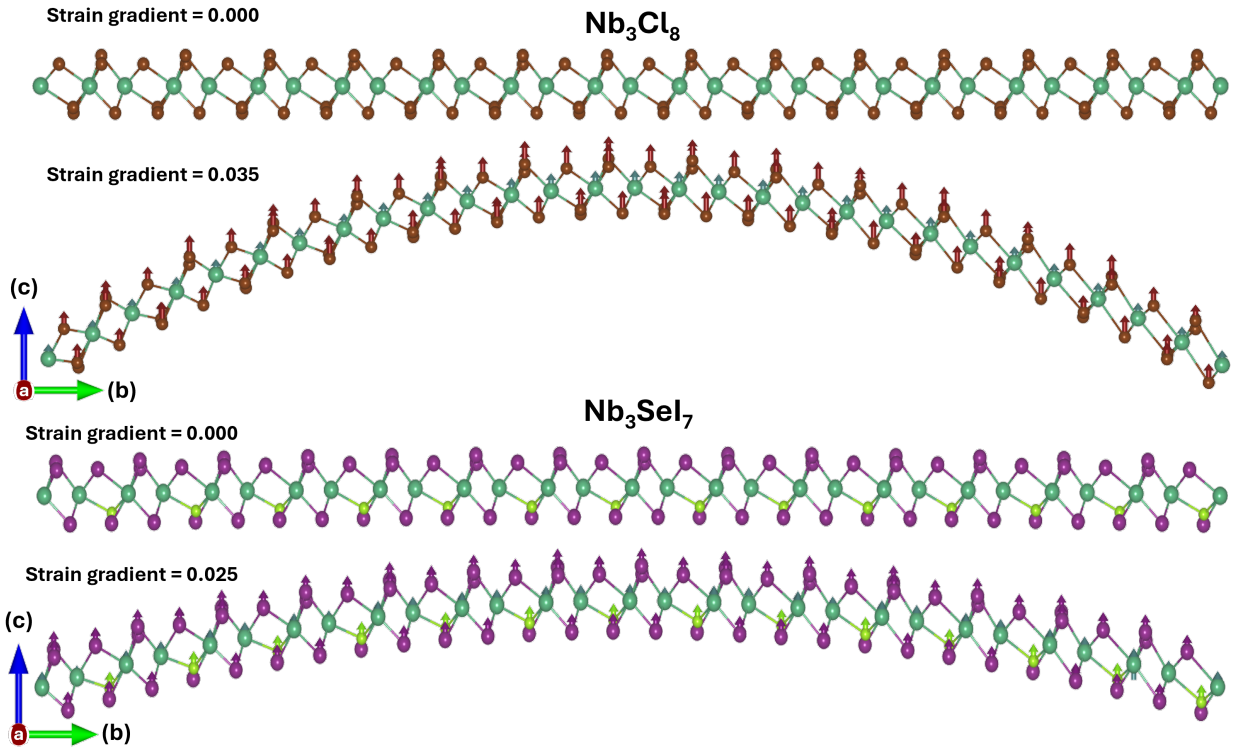


Fig. 6. Atomic configurations illustrating the bending-induced out-of-plane dipole response in representative layered kagome monolayers. The upper panels show the undeformed and bent configurations of a non-Janus Nb₃Cl₈-type system, while the lower panels show the corresponding configurations of a Janus Nb₃SeI₇-type system. The arrows indicate the local atomic dipole component along the out-of-plane direction, p_z . In the flat configuration, the dipoles are weak or partially compensated. Under bending deformation, the nonuniform strain field breaks local symmetry and induces a finite p_z , confirming the flexoelectric origin of the curvature-induced polarization. The stronger and more coherent dipole response in the Janus structure originates from the combined effect of intrinsic out-of-plane polarity, asymmetric chalcogen–halogen bonding, lattice softness, and curvature-induced charge redistribution.

Figure 6 shows the initial and bent configurations of representative non-Janus Nb₃Cl₈ and Janus Nb₃SeI₇ layered kagome monolayers. The arrows indicate the local atomic dipole component along the out-of-plane direction, p_z . In the undeformed configuration, the atomic layers remain nearly flat and the local dipoles are either weak or partially compensated by the opposite atomic environment. After bending, a clear upward p_z component is generated throughout the curved

monolayer, confirming that the applied curvature creates a nonuniform strain field and activates a strain-gradient-induced polarization response. For the non-Janus Nb_3Cl_8 system (see Fig. 6(a)), the top and bottom halogen environments are chemically similar. Therefore, in the flat configuration, the local dipoles associated with Nb–X bonds tend to compensate each other across the two sides of the monolayer. Under bending deformation, however, this compensation is no longer complete. The outer side of the bent sheet experiences tensile strain, whereas the inner side experiences compressive strain. This asymmetric deformation changes the Nb–X bond lengths, modifies the local charge distribution, and generates a finite out-of-plane dipole moment. Hence, the observed upward arrows in the bent non-Janus structure mainly originate from flexoelectric polarization rather than from intrinsic Janus polarity (see Fig. 6(a)).

In contrast, the Janus M_3YZ_7 monolayer already possesses chemical asymmetry because the two surfaces are terminated by different chalcogen and halogen atoms (see Fig. 6(b)). This broken out-of-plane mirror symmetry creates an intrinsic polar environment even before bending. When curvature is applied, the strain gradient further enhances the separation of positive and negative charge centers, leading to a stronger and more spatially coherent p_z response. The larger local dipoles in Janus systems therefore arise from the cooperative effect of intrinsic structural polarity and curvature-induced charge redistribution. The magnitude of the local p_z response is governed by several competing factors. Materials containing heavier and more polarizable halogens, such as Br or I, generally exhibit larger bending-induced dipoles because their diffuse p -orbitals are more easily distorted under a local electric field (see Fig. 5(b)). At the similar time, longer and softer metal–halogen bonds allow larger curvature-induced internal displacement, which further enhances the dipole moment. This explains why Br- and I-containing kagome monolayers usually show stronger flexoelectric responses than Cl-containing systems. However, the dipole magnitude is not controlled by polarizability alone. Excessive dielectric screening in highly polarizable systems can reduce the effective local electric field, so the final response depends on the balance between atomic polarizability, bond softness, charge-transfer strength, and local dielectric screening.

For Janus chalcogenide-halides, the chalcogen atom also plays an important role. A larger chalcogen atom such as Te increases the contribution of electric-field and lattice softness, which can enhance curvature-induced structural relaxation. However, stronger dielectric screening may partially suppress the local electric field. In contrast, lighter chalcogens such as S can form shorter and stronger metal–chalcogen bonds, producing larger bond-force constants and more efficient strain-gradient-induced charge redistribution. Therefore, the largest p_z response is obtained when Janus polarity, lattice softness, and local electric-field enhancement act together without excessive screening. The spatial variation of the arrows in Fig. 6 also reflects the nonuniform nature of bending deformation. The dipole response is stronger in regions of larger curvature because the strain gradient is larger there. Near regions with weaker curvature, the local p_z arrows become smaller, indicating a reduced flexoelectric contribution. This confirms that the generated out-of-plane polarization is directly linked to the local strain gradient rather than to a uniform tensile or compressive strain. Therefore, Fig. 6 provides an atomistic visualization of the microscopic origin of flexoelectricity in layered kagome monolayers: bending breaks the local symmetry, redistributes charge between metal and chalcogen/halogen atoms, and produces a measurable out-of-plane dipole moment.

4. Conclusions

In summary, we systematically investigated the optical, thermodynamic, mechanical, piezoelectric, and flexoelectric responses of non-Janus Nb_3X_8 ($X = \text{Cl}, \text{Br}, \text{I}$) and Janus M_3YZ_7 ($M = \text{Nb}, \text{Ta}$; $Y = \text{S}, \text{Se}, \text{Te}$; $Z = \text{Cl}, \text{Br}, \text{I}$) layered kagome monolayers using first-principles DFT calculations and MD simulations based on trained MTPs combined with a QP model. These are the major finding as follows:

- The optical spectra show strong polarization anisotropy, with a much larger in-plane ($E \parallel b$) response than the out-of-plane ($E \parallel c$) response, confirming that light-matter interaction is dominated by the Nb-based breathing-kagome plane. For non-Janus Nb_3X_8 , the optical strength follows $\text{Nb}_3\text{I}_8 > \text{Nb}_3\text{Br}_8 > \text{Nb}_3\text{Cl}_8$, while the absorption coefficient reaches $\sim 10^6 \text{ cm}^{-1}$, indicating strong visible and near-ultraviolet absorption. Among Janus Nb_3YZ_7 , Nb_3SeI_7 shows the strongest peak response, whereas Nb_3TeI_7 provides broader absorption due to diffuse Te- p states and spin-orbit-induced band mixing.
- The pressure-dependent thermodynamic properties reveal that Nb_3I_8 has the largest heat capacity and entropy, followed by Nb_3Br_8 and Nb_3Cl_8 , owing to heavier halogen mass and softer low-frequency phonon modes. The lattice thermal conductivity decreases with increasing temperature for all Nb_3X_8 compounds, confirming that anharmonic phonon-phonon scattering dominates thermal transport.
- The mechanical response follows the stiffness order $\text{Nb}_3\text{Cl}_8 > \text{Nb}_3\text{Br}_8 > \text{Nb}_3\text{I}_8$. Specifically, Nb_3Cl_8 is approximately 1.10 and 1.27 times stiffer than Nb_3Br_8 and Nb_3I_8 , respectively, due to shorter Nb-Cl bonds, stronger force constants, and reduced lattice deformability. The trained MTPs accurately reproduce the DFT stiffness trends, validating their use for large-scale elastic deformation simulations.
- Janus M_3YZ_7 monolayers exhibit stronger piezoelectricity than non-Janus Nb_3X_8 because broken out-of-plane mirror symmetry, intrinsic polarity, and asymmetric charge redistribution enhance strain-induced polarization. In particular, Nb_3SeI_7 shows piezoelectric coefficients approximately 1.40 and 1.26 times larger than those of Nb_3SeBr_7 and Nb_3SeCl_7 , respectively, demonstrating the combined role of local-field variation, charge transfer, dielectric screening, and internal relaxation.
- Both non-Janus and Janus kagome monolayers generate out-of-plane polarization under bending-induced strain gradients, with flexoelectric coefficients of $\sim 0.3\text{--}0.8 \text{ nC/m}$. Janus I-containing compounds generally show the largest responses because structural polarity, large atomic polarizability, lattice softness, and curvature-induced charge redistribution act cooperatively, whereas Cl-containing and some Ta-based systems exhibit smaller responses due to stronger bonding rigidity and weaker local-field variation.

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